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Vane forward rail for gas turbine engine assembly

Abstract

Vane assemblies for gas turbine engines are described. The vane assemblies include a platform having an interior platform surface, a forward rail, and an aft rail defining a plenum. An airfoil extends radially inward from the platform on a side opposite the forward and aft rails and includes a leading edge cavity that is open at the platform. A platform feed structure is arranged on the platform about the leading edge cavity and in the plenum and defines a fluid path through the forward rail and into the leading edge cavity. A cover plate is arranged on a top surface of the platform feed structure and configured to fluidly separate the plenum of the platform from the leading edge cavity and define a turning plenum. The cover plate defines a turning contour surface that is shaped to turn an airflow from an axial flow direction to a radial flow direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application Ser. No. 63/304,198, filed Jan. 28, 2022, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND
(1) The subject matter disclosed herein generally relates to gas turbine engine assemblies and, more particularly, to vane and rail assemblies associated with cooling schemes for airfoils of gas turbine engines.
(2) A gas turbine engine assembly may include one or more vane assemblies, each vane in a respective vane assembly having one or more vane rails. An amount of air flow available to cores or cavities of airfoils accessible via the vane rail varies based on size, shape, and location of the cores or cavities. Airflow access to the cores or cavities is typically provided through drilled channels. Current designs of cores and cavities are configured such that access to vane rail portions to drill to the cores or cavities is difficult and does not necessarily promote efficient air flow.
BRIEF DESCRIPTION
(3) According to some embodiments, vane assemblies for gas turbine engines are provided. The vane assemblies include a platform having an interior platform surface, a forward rail, and an aft rail, wherein the interior platform surface, the forward rail, and the aft rail define a plenum, an

airfoil extending radially inward from the platform on a side opposite the forward and aft rails, the airfoil having a leading edge cavity that is open at the platform, a platform feed structure arranged on the platform about the leading edge cavity and in the plenum and defining a fluid path through the forward rail and into the leading edge cavity of the airfoil, and a cover plate arranged on a top surface of the platform feed structure, the cover plate configured to fluidly separate the plenum of the platform from the leading edge cavity and define a turning plenum, wherein the cover plate comprises a turning contour surface that is shaped to turn an airflow from an axial flow direction to a radial flow direction within the turning plenum.

(4) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the platform feed structure has an open top that is defined by a top surface and the cover plate is attached to the top surface.

(5) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the cover plate is attached to the top surface by one of welding, bonding, and adhesive.

(6) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the turning contour defines a dome shape on the cover plate.

(7) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the turning contour defines an apex on the cover plate that is aligned with an inlet and an outlet of the platform feed structure.

(8) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the outlet is aligned with the leading edge cavity of the airfoil.

(9) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the top surface is substantially planar extending from a forward position on an interior rail surface of the forward rail and an aft position on the interior platform surface of the platform.

(10) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the top surface defines a contoured surface extending from a forward position on an interior rail surface of the forward rail and an aft position on the interior platform surface of the platform.

(11) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the contoured surface is convex.

(12) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the contoured surface is concave.

(13) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the platform feed structure comprises an inlet in the forward rail and an outlet in the platform.

(14) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the platform feed structure defines a first feed path through the platform feed structure from the forward rail to the leading edge cavity and a second feed path through the platform feed structure from the forward rail to a platform cooling cavity of the platform.

(15) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the first feed path is defined through a first extension of the platform feed structure and the second feed path is defined through a second extension of the platform feed structure, wherein the cover plate is positioned over a portion of at least the first feed path.

(16) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the first extension and the second extension

have different heights relative to the interior platform surface.
(17) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the platform feed structure extends at least 20% of an axial distance between the forward rail and the aft rail.
(18) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the platform feed structure extends between 20% and 70% of an axial distance between the forward rail and the aft rail.
(19) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the platform feed structure extends less than 20% of an axial distance between the forward rail and the aft rail.
(20) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the airfoil comprises an aft cavity arranged aft of the leading edge cavity, wherein the aft cavity is fluidly connected to the plenum of the platform.
(21) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the platform is an outer diameter platform and the airfoil extends radially inward to an inner diameter platform.
(22) In addition to one or more of the features described herein, or as an alternative, further embodiments of the vane assemblies may include that the platform feed structure is sized to allow a baffle to pass therethrough to install the baffle within the leading edge cavity of the airfoil.
(23) The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated otherwise. These features and elements as well as the operation thereof will become more apparent in light of the following description and the accompanying drawings. It should be understood, however, that the following description and drawings are intended to be illustrative and explanatory in nature and non-limiting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:
(2) FIG. **1** is a schematic illustration of a gas turbine engine that may employ embodiments of the present disclosure;
(3) FIG. **2** is a schematic illustration of an engine section of a gas turbine engine that may incorporate embodiments of the present disclosure;
(4) FIG. **3** is a schematic illustration of a vane assembly that may incorporate embodiments of the present disclosure;
(5) FIG. **4**A is a schematic illustration of a vane assembly in accordance with an embodiment of the present disclosure;
(6) FIG. **4**B is a side schematic illustration of a portion of the vane assembly of FIG. **4**A;
(7) FIG. **4**C is top-down (radially inward) view of the vane assembly of FIG. **4**A;
(8) FIG. **5** is a schematic illustration of a vane assembly in accordance with an embodiment of the present disclosure;
(9) FIG. **6** is a schematic illustration of a vane assembly in accordance with an embodiment of the present disclosure;
(10) FIG. **7**A is a schematic illustration of a vane assembly in accordance with an embodiment of the present disclosure including cover plates;
(11) FIG. **7**B is a schematic illustration of a cover plate of FIG. **7**A;
(12) FIG. **7**C is a schematic illustration of an alternate view of the cover plate of FIG. **7**B;

(13) FIG. **8**A is a schematic illustration of a vane assembly in accordance with an embodiment of the present disclosure;
(14) FIG. **8**B is a front elevation (looking aft) of the vane assembly of FIG. **8**A; and
(15) FIG. **9** is a schematic illustration of a vane assembly in accordance with an embodiment of the present disclosure including cover plates.

DETAILED DESCRIPTION

(16) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures. As shown and described herein, various features of the disclosure will be presented. Various embodiments may have the same or similar features and thus the same or similar features may be labeled with the same reference numeral, but preceded by a different first number indicating the figure to which the feature is shown. Although similar reference numbers may be used in a generic sense, various embodiments will be described and various features may include changes, alterations, modifications, etc. as will be appreciated by those of skill in the art, whether explicitly described or otherwise would be appreciated by those of skill in the art.

(17) FIG. **1** schematically illustrates a gas turbine engine **100**. The gas turbine engine **100** is disclosed herein as a two-spool turbofan that generally incorporates a fan section **102**, a compressor section **104**, a combustor section **106**, and a turbine section **108**. FIG. **1** is one example of a gas turbine engine that may incorporate embodiments of the present disclosure. However, alternative engine configurations that may incorporate embodiments described herein may include other systems, features, and/or arrangement of components, as will be appreciated by those of skill in the art. The fan section **102** is configured to drive air along a bypass flow path B in a bypass duct, while the compressor section **104** is configured to drive air along a core flow path C for compression and communication into the combustor section **106** then expansion through the turbine section **108**. Although depicted as a two-spool turbofan gas turbine engine in the disclosed non-limiting embodiment, it should be understood that the concepts described herein are not limited to use with two-spool turbofans as the teachings may be applied to other types of turbine engines without departing from the scope of the present disclosure.

(18) The gas turbine engine **100** generally includes a low speed spool **110** and a high speed spool **112**, each spool mounted for rotation about an engine central longitudinal axis A relative to an engine static structure **114** via one or more bearing systems **116**. It should be understood that various bearing systems **116** at various locations may alternatively or additionally be provided, and the location of such bearing systems may be varied as appropriate to the application and/or engine configuration.

(19) The low speed spool **110** generally includes an inner shaft **118** that interconnects a fan **120**, a low pressure compressor **122**, and a low pressure turbine **124**. The inner shaft **118** is connected to the fan **120** through a speed change mechanism, which in the configuration of the gas turbine engine **100** shown in FIG. **1** is illustrated as a geared architecture **126** to drive the fan **120** at a lower speed than the low speed spool **110**. The high speed spool **112** includes an outer shaft **128** that interconnects a high pressure compressor **130** and high pressure turbine **132**. A combustor **134** is arranged in the gas turbine engine **100** between the high pressure compressor **130** and the high pressure turbine **132**. The engine static structure **114** is configured to support the bearing systems **116**. The inner shaft **118** and the outer shaft **128** are concentric and rotate via the bearing systems **116** about the engine central longitudinal axis A which is collinear with their longitudinal axes.

(20) In operation, the core airflow in the core airflow path C is compressed by the low pressure compressor **122**, then the high pressure compressor **130**, mixed and burned with fuel in the combustor **134**, then expanded over the high pressure turbine **132**, and finally the low pressure turbine **124**. The turbines **134**, **124** may be configured to rotationally drive the low speed spool **110** and the high speed spool **112**, respectively, in response to the expansion of the core airflow along the core airflow path C. It will be appreciated that each of the positions of the fan section **102**, the

compressor section **104**, the combustor section **106**, the turbine section **108**, and the fan drive gear system (e.g., geared architecture **126**) may be varied relative to each other. For example, in some non-limiting configurations, the geared architecture **126** may be located aft of the combustor section **106** or even aft of the turbine section **108**, and the fan section **102** may be positioned forward or aft of the location of the geared architecture **126**.

(21) The gas turbine engine **100**, in one non-limiting example, is a high-bypass geared aircraft engine. In some examples, a bypass ratio of the gas turbine engine **100** may be greater than about six (6), with an example embodiment being greater than about ten (10). In some embodiments, the geared architecture **126** may be an epicyclic gear train, such as a planetary gear system or other gear system, with a gear reduction ratio of greater than about 2.3 (2.3:1). In some embodiments, the low pressure turbine **124** may have a pressure ratio that is greater than about five (5). In one embodiment, a bypass ratio of the gas turbine engine **100** may be greater than about ten (10:1). In some embodiments, a diameter of the fan **120** may be significantly larger than that of the low pressure compressor **122**. In some embodiments, the low pressure turbine **124** may have a pressure ratio that is greater than about five (**5**:**1**). A pressure ratio of the low pressure turbine **124** is a pressure measured prior to an inlet of low pressure turbine **124** as related to a pressure at the outlet of the low pressure turbine **124** prior to an exhaust nozzle or other downstream component. It should be understood that the above parameters are only exemplary of one embodiment of a geared architecture engine and that the present disclosure is applicable to other gas turbine engines including, but not limited to, direct drive turbofans.

(22) In some embodiments, a significant amount of thrust is provided by the bypass flow B due to a high bypass ratio. The fan section **102** of the gas turbine engine **100** may be designed for particular flight condition(s)—typically cruise at about 0.8 Mach and about 35,000 feet (10,688 meters). The flight condition of 0.8 Mach and 35,000 ft (10,688 meters), with the engine at its best fuel consumption—also known as “bucket cruise Thrust Specific Fuel Consumption (‘TSFC’)”—is the industry standard parameter of lbm of fuel being burned divided by lbf of thrust the engine produces at that minimum point. “Low fan pressure ratio” is the pressure ratio across the fan blade alone, without a Fan Exit Guide Vane (“FEGV”) system. The low fan pressure ratio as disclosed herein according to one non-limiting embodiment is less than about 1.45. “Low corrected fan tip speed” is the actual fan tip speed in ft/sec divided by an industry standard temperature correction of [(Tram ° R)/(518.7° R)].sup.0.5. The “Low corrected fan tip speed” as disclosed herein according to one non-limiting embodiment is less than about 1150 ft/second (350.5 m/sec). the gas turbine engine **100** of FIG. **1** may be configured with one or more components as shown and described herein.

(23) Referring now to FIG. **2**, a schematic illustration of an engine section **200** of a gas turbine engine that can incorporate embodiments of the present disclosure is shown. The engine section **200** shown in FIG. **2** may be illustrative of a portion of a turbine section or a compressor section of a gas turbine engine, such as shown and described above with respect to FIG. **1**. The engine section **200** includes a first stage vane **202** and a second stage vane **204**. The first stage vane **202** is located forward of a first one of a pair of turbine disks **206**. Each of the turbine disks **206** includes a plurality of turbine blades **208** secured thereto. The turbine blades **208** are configured to rotate proximate to blade outer air seals **210** at tips thereof In this illustrative configuration. the blade outer air seals **210** are located aft of the first stage vane **202**. The second stage vane **204** is located between the pair of turbine disks **206**.

(24) In one non-limiting example, the first stage vane **202** is the first vane of a high pressure turbine section that is located aft of a combustor section (see., e.g., FIG. **1**). The second stage vane **204** is located aft of the first stage vane **202** between the pair of turbine disks **206**. The second stage vane **204** and the first stage vane **202** are located circumferentially about an engine central longitudinal axis A to provide a stator assembly **212**. Hot gases from the combustor section are configured to flow through the engine section **200** in a direction of arrow **214**, thus the hot gases

first interact with the first stage vane **202** and then subsequently with the second stage vane **204**, in a flow path direction. Although a two-stage (e.g., two vane sections) system is illustrated, other engine sections/configurations are considered to be within the scope of various embodiments of the present disclosure.
(25) Referring now to FIG. **3**, a side view of a portion of an example of a vane assembly **300** that may be implemented within gas turbine engines is schematically shown. The vane assembly **300**, as shown, includes an airfoil **302** disposed between an inner diameter platform **304** and an outer diameter platform **306**. The vane assembly **300** may include a vane forward rail **308** that is configured to engage with other parts of an engine assembly (e.g., hooks or the like). The vane assembly **300** may be located within a compressor section or a turbine section as described above. The airfoil **302** extends in a radial direction (relative to an engine axis). Cooling air for the airfoil **302** may be supplied into one or more internal cavities of the airfoil **302**. The cooling air may be directed into the internal cavities of the airfoil **302** from an inner diameter plenum **310** or an outer diameter plenum **312**. In some configurations and as shown, a cooling airflow **314** may be directed through the forward rail **308** and into the internal cavities of the airfoil **302** through a platform feed structure **316**. The platform feed structure **316** may be configured to receive the cooling airflow **314** from an upstream location, pass the cooling airflow **314** through the forward rail **308**, turn such airflow **314** within the platform feed structure **316**, and into one or more internal cavities of the airfoil **302** (e.g., leading edge and midbody cavities) and/or into internal cavities and channels within the outer diameter platform **306** to provide cooling thereto.
(26) Due to airfoil sizes and/or other engine system considerations, a flow allotment and pressure level consistent with existing configurations may not be sufficient to cool next generation designs using traditional leading edge cooling configurations (e.g., discrete leading edge cavities and/or impingement baffles). Additionally, structural concerns may require maximizing a distance between a leading edge of the airfoil and a first rib within the airfoil (e.g., defining a leading edge cavity of the airfoil). When implemented, the cooling airflow **314** will enter the platform area (e.g., at the vane forward rail **308**) perpendicular to the airfoil cavity (e.g., left-to-right on FIG. **3**). As such, the airflow **314** will have to be directed or turned downward (e.g., radially inward) into at least one cooling cavity or cavities of the airfoil **302**. In certain situations, some airfoil cooling configurations may require a feed system that allows for a baffle to be inserted into the airfoil cavity with a perpendicular feed. Furthermore, consideration of the platform feed structure **316** must also leave enough clearance in the outer diameter plenum **312** to allow turbine cooling and leakage air (TCLA) cooling air pipes to pressurize the outer diameter plenum **312** to the required supply pressure necessary to provide adequate cooling air mass flow to all platform and airfoil cooling circuits in order to meet thermal performance and durability life requirements. Additionally, debris in current/existing configurations has demonstrated plugged feeds, which can starve the entire leading edge circuit of the airfoil, which in turn can lead to premature platform and airfoil oxidation, thermal mechanical fatigue, and creep distress failure modes.
(27) In accordance with embodiments of the present disclosure, platform feed structures are provided to extend from a platform surface to a feed radius location. Such structures may extend in an aftward direction and back toward a trailing edge rail to allow the option for a leading edge baffle to be installed within the airfoil of the vane assembly. The structures described herein are designed such that a baffle may be inserted through the outer diameter platform of the airfoil assembly and into the interior of the airfoil structure. In some configurations, the airfoil has a curvature in the circumferential direction (i.e., defining the airfoil surfaces), which in turn requires the installed baffle to be curved both tangentially and/or axially as well. This curvature can result in an upper limit of the platform feed structures before the structure itself will interfere with installation of the baffle. By minimizing the structural size of the platform feed structures, a reduction of cross-sectional area within the platform cooling area and minimizing the amount of material present for both structural and weight benefits may be achieved. The platform feed

structures may have axial lengths (e.g., extending aft from a forward rail) that are selected to accommodate the installation of a baffle through the structure and into an airfoil. It will be appreciated that the tangential and axial bow of a vane aerodynamic geometry may limit the radial height and an axial extent of an outer diameter leading edge, mid body, and/or trailing edge radially oriented feed cavity structure, due to interference between the outer diameter feed cavity structure and a bowed baffle geometry when inserted into the airfoil during the vane assembly process.

(28) Further, in accordance with some embodiments of the present disclosure, the platform feed structure may be shaped to angle or otherwise direct a flow downward (e.g., radially inward) into a leading edge circuit of the airfoil. In accordance with some embodiments, the platform feed structures of the present disclosure may be sized and shaped to ensure that an outer diameter plenum projected cross-sectional area is not disrupted significantly or negatively that may increase circumferential pressure losses within the outer diameter plenum that is configured to supply cooling air to the entire annulus of the vane assembly from discrete circumferential locations. An axial length (i.e., a direction along an engine axis) extends from a leading edge rail toward a trailing rail of the platform for a distance large enough to enable installation of a baffle into the airfoil of the vane assembly. Such axial length or distance can also accommodate turning of a cooling flow 90° without significant losses.

(29) To minimize pressure losses associated with a 90° turn from the axial feed inlet to the radially extending airfoil cooling cavities, it may be desirable to minimize flow separation that naturally occurs when cooling airflow is turned 90° immediately upstream the airfoil outer diameter inlet cooling feed cavities. It may also be desirable to ensure cooling mass flow uniformity along the axial and circumferential inlet flow plane (or outer diameter airfoil inlet feed) which is oriented nearly parallel to the centerline of the engine axis and is coincident with the outer diameter of the airfoil platform intersection. Ensuring mass flow velocity and pressure uniformity at the inlet of the outer diameter of the airfoil cooling cavities may reduce airfoil feed inlet pressure losses, improve airfoil cavity fill characteristics, and/or increase local convective cooling performance along the outer diameter airfoil-platform fillet region.

(30) In some embodiments, the turning of the flow may be controlled or dictated by a shaped or contoured cover plate that may be formed to capture or catch the axial cooling flow fed through inlet feed apertures in the leading edge forward rail structure and turn the flow (passively) along a contoured surface of the cover plate. Such control may mitigate an axial velocity flow jetting and flow separation that would naturally occur with cooling flow that is moving axially and required to turn 90° to onboard into the radially oriented vane turbine airfoil inlet. Further, in addition to creating a plenum and turning airflow, the platform feed structures allow the use of a relatively large slot feed aperture on the vane forward rail (e.g., vane forward rail **308**). Incorporation of larger cross-sectional feed apertures may permit debris in the cooling air feed system to pass through without clogging the feed aperture/slot. Additionally, such relatively large slots/openings can minimize inlet pressure losses, allowing the use of more pressure within the airfoil for cooling. Prior configurations were subject to significant pressure losses at the inlet, which some embodiments and configurations described herein address by reducing or eliminating such losses.

(31) Turning now to FIGS. **4**A-**4**C, schematic illustrations of a vane assembly **400** in accordance with an embodiment of the present disclosure are shown. The vane assembly **400** includes an airfoil **402** disposed between an outer diameter platform **404** and an inner diameter platform **406**. The airfoil **402** has a leading edge **408** and a trailing edge **410**. The airfoil **402**, in this embodiment, includes a leading edge cavity **412** and a trailing edge cavity **414** that are configured to receive cooling air and provide cooling to the airfoil **402**. In this embodiment, a baffle **416** is arranged within the leading edge cavity **412** to serve as a space-eater component and to receive and direct cooling air onto internal surfaces of the airfoil **402**. The airfoil **402** may be integrally formed with the outer diameter platform **404** and the inner diameter platform **406** to form the vane assembly **400**. The outer diameter platform **404** may include a platform cooling cavity **418** which is arranged

to cool portions of the outer diameter platform **404** exposed to hot gaspath air. Although not expressly shown, the trailing edge cavity can include a baffle and/or may be arranged as a serpentine flow path, as will be appreciated by those of skill in the art. Further, the inner diameter platform **406** may include internal cavities for cooling thereof, as will be appreciated by those of skill in the art.

(32) As shown in FIGS. **4**A-**4**C, the outer diameter platform **404** includes a forward rail **420** and an aft rail **422**. An outer diameter plenum **424** is defined between the forward rail **420** and an aft rail **422**. The outer diameter platform **404** includes a platform feed structure **426** formed as part of the outer diameter platform **404** and positioned within the outer diameter plenum **424**. The platform feed structure **426** is a structure of the outer diameter platform **404** that has one or more feed paths defined therethrough. A first feed path **428** extends between an inlet **430** in the forward rail **420** and an outlet **432** that directs cooling air into the baffle **416**. A second feed path **434** extends between an inlet **436** in the forward rail **420** and an outlet **438** that directs cooling air into the platform cooling cavity **418**.

(33) Although prior platform feed structures have existed, the platform feed structure **426** of the present embodiment is significantly enlarged as compared to such prior configurations. For example, in an axial direction (e.g., in a direction between the forward rail **420** and the aft rail **422**), the length or extent of the platform feed structure **426** may be sized to allow for the installation of a baffle within the airfoil (vane) extending from the platform. Further, for example, in a circumferential direction (e.g., pressure-to-suction direction relative to the airfoil) the size may be increased relative to prior configurations to enable an enlarged feed slot to reduce plugging and/or pressure losses.

(34) For example, and in accordance with some embodiments of the present disclosure and without limitation, an axial length L.sub.A of the platform feed structure **426** may extend at least 20% of the axial distance L.sub.P of the platform **404** between the forward rail **420** and the aft rail **422**. In some non-limiting configurations, the axial length L.sub.A may be between 20% and 70% of the axial distance L.sub.P of the platform **404**. This increased distance (e.g., as compared to prior configurations having a profile in the axial direction of less than 10%) allows for improved turning of cooling flow that must change from an axial flow direction to a radial flow direction (e.g., 90° turn) as the cooling flow enters the outer diameter airfoil inlet feed cavities. The platform feed structure **426** is sized to accommodate installation of a baffle into the airfoil and may enable a smoother fluid transition from an axial flow direction to a radial flow direction while also minimizing impacts on crossflow through the outer diameter plenum **424**, and increased pressure losses, and/or plugging at the leading-edge rail inlet feed apertures **430**, **436**. The axial extent may also be controlled, at least in part, based on the size and shape of the baffle to be installed within the airfoil.

(35) Because the platform feed structure **426** extends for a larger axial distance than prior configurations, to minimize impacts on crossflow through the outer diameter plenum **424**, a top surface **440** of the platform feed structure **426** may be angled from a forward position **442** on the forward rail **420** to an aft position **444** on the outer diameter platform **404**. The top surface **440** may be substantially linear from the forward position **442** to the aft position **444**. However, in other embodiments, the top surface **440** may have a curvature or non-linear slope from the forward position **442** to the aft position **444**, such as resulting in a concave or convex surface.

(36) As noted, advantages of the increased axial length include the ability to have an airfoil with a baffle installed therein. In conventional configurations, the airfoil feed inlet cross-sectional area openings and limitations associated with the platform radially extending feed structures may be insufficient to allow for baffle installation. That is, the hole sizes are too small to accommodate a baffle to be installed therethrough and into the interior cavity of the airfoil. However, in contrast, the platform feed structure **426** of the present disclosure has a large enough size to accommodate the insertion of the baffle **416** through the opening defined by the outlet **432** and into the leading

edge cavity **412** of the airfoil **402**. That is, the outlet **432** of the platform feed structure **426** is sized to allow for the baffle **416** to pass therethrough to be installed within the leading edge cavity **412** of the airfoil **402**.

(37) Embodiments of the present disclosure are directed to enlarged platform feed structures (e.g., smokestacks) that allow for the insertion of a baffle for more complex cooling circuits within airfoils of vane assemblies. The platform feed structures have a forward end that mates up against the front rail of the platform, where the cooling air is introduced. This allows the feed slot (inlet) to be enlarged to a width (circumferential direction) of the platform feed structure or at least associated inlet (e.g., inlet **430**). For example, as compared to prior configurations, the inlet area may be doubled in size (cross-sectional area) to prevent particle accumulation. In one non-limiting example, the cross-sectional area at the inlet (e.g., inlet **430**) may be about 0.025 sq-in or greater, such as about 0.03 sq-in, about 0.04 sq-in, about 0.05 sq-in, or greater. It will be appreciated that the aforementioned cross-sectional areas are merely for example and are not to be limiting on the size of such inlet areas, but rather provide some examples and information regarding such larger size inlets as compared to prior configurations. The platform feed structure can be extended in the circumferential direction (e.g., to allow larger feed area to minimize pressure loss and plugging risk) and in the axial direction (e.g., to allow a circuit to cover more or less airfoil surface area depending on cooling requirements).

(38) Turning now to FIG. **5**, a schematic illustration of a vane assembly **500** in accordance with an embodiment of the present disclosure is shown. The vane assembly **500** may be similar to that shown and described above. For example, the vane assembly **500** includes an outer diameter platform **502** and an airfoil **504** extending radially inward from the outer diameter platform **502**. The outer diameter platform **502** includes a forward leading edge rail **506** having an interior rail surface **508** and an interior platform surface **510**. The interior rail surface **508** and the interior platform surface **510** define, in part, an outer diameter plenum, similar to that shown and described above.

(39) The vane assembly **500** includes a platform feed structure **512**. The platform feed structure **512** is configured to define a flow path for cooling air that passes through the forward leading edge rail **506** and into an interior of, at least, the airfoil **504** (specifically a baffle installed within the airfoil **504**). The platform feed structure **512** has a top surface **514** that defines a turning plenum **516** therein. The platform feed structure **512** is a structure that is integrally formed with the material of the outer diameter platform **502**. Cooling air will enter into the turning plenum **516** from an inlet aperture **518** and is turned within the platform feed structure **512** from an axial flow to a radial flow. The turned flow will be directed into an airfoil body, and in some embodiments, into a baffle that is arranged and positioned within an airfoil body (e.g., within a cavity of the airfoil **504**). Although not shown, the platform feed structure **512** may include a separate (second) inlet for receiving cooling air and directing such air into one or more cooling cavities of the outer diameter platform **502**.

(40) The platform feed structure **512** has an open top **519** that is defined by the top surface **514** and the turning plenum **516**. The open top **519** is sized and shaped to permit a baffle to pass through the platform feed structure **512** and to be installed within an internal cavity of the airfoil **504** (e.g., a leading edge cavity). In some embodiments, the open top **519** (e.g., open area at the top of the radially oriented platform feed structure **512**) may be capped or covered with a cover plate (not shown). The top surface **514**, in this embodiment, is substantially planar and angled, which provides advantages to producibility (e.g., installation of the cover plate, pulling a cast, etc.). As such, the top surface **514** is defined within a plane that extends through a forward position **520** on the interior rail surface **508** of the forward leading edge rail **506** and an aft position **522** on the interior platform surface **510** of the platform **502**.

(41) Turning now to FIG. **6**, a schematic illustration of a vane assembly **600** in accordance with an embodiment of the present disclosure is shown. The vane assembly **600** may be similar to that

shown and described above. For example, the vane assembly **600** includes an outer diameter platform **602** and an airfoil **604** extending radially inward from the outer diameter platform **602**. The outer diameter platform **602** includes a forward rail **606** having an interior rail surface **608** and an interior platform surface **610**. The interior rail surface **608** and the interior platform surface **610** define, in part, an outer diameter plenum, similar to that shown and described above.

(42) The vane assembly **600** includes a radially oriented platform feed structure **612**. The platform feed structure **612** is configured to define a flow path for cooling air that passes through the forward rail **606** and into an interior of, at least, the airfoil **604** (specifically a baffle installed within the airfoil **504**). The platform feed structure **612** has a top surface **614** that defines a turning plenum **616** therein. The platform feed structure **612** is a structure that is integrally formed with the material of the outer diameter platform **602**. Cooling air will enter into the turning plenum **616** from an inlet **618** and is turned within the platform feed structure **612** from an axial flow to a radial flow. The turned flow will be directed into an airfoil body, and in some embodiments, into a baffle that is arranged and positioned within an airfoil body (e.g., within a cavity of the airfoil **604**). Although not shown, the platform feed structure **612** may include a separate (second) inlet for receiving cooling air and directing such air into one or more cooling cavities of the outer diameter platform **602**.

(43) The platform feed structure **612** has an open top **620** that is defined by the top surface **614** and the turning plenum **616**. The open top **620** is sized and shaped to permit a baffle to pass through the platform feed structure **612** and to be installed within an internal cavity of the airfoil **604** (e.g., a leading edge cavity). In some embodiments, the open top **620** (e.g., open area at the top of the radially oriented platform feed structure **612**) may be capped or covered with a cover plate (not shown). The top surface **614**, in this embodiment, is curved or contoured. That is, the top surface **614** is defined by a curved plane that extends through a forward position **620** on the interior rail surface **608** of the forward rail **606** and an aft position **622** on the interior platform surface **610** of the platform **602**. The resulting top surface **614** is a substantially concave surface that the cross-sectional area of the platform feed structure **612** is less than a cross-sectional area of the platform feed structure **512** of FIG. **5**. It will be appreciated that the top surface of the radially oriented platform feed cavity may also incorporate a convex surface opposite of the curvature illustrated in FIG. **6**, without departing from the scope of the present disclosure. In such embodiments, the top surface of the platform feed structure may be convex and bow outward (e.g., radially outward) as compared to the bowing radially inward as illustrated in FIG. **6**. The selection of this geometry may be based, in part, on the amount of required space within the turning plenum of the platform feed structure, weight considerations, material considerations, and/or considerations related to optimizing structural concerns related to the added stiffness provided by the platform feed structure. A convex platform feed structure will have the largest internal volume of the turning plenum, as compared to a constant slope (e.g., FIG. **5**) or a concave slope (e.g., FIG. **6**).

(44) As noted above, the top of the platform feed structure, in accordance with embodiments of the present disclosure, may be covered by a cover plate. Such cover plates can fluidly separate the turning plenum from the outer diameter plenum of the platform. The cover plate is configured to be installed to the top surface of the platform feed structure. In some embodiments, the cover plate may be a substantial flat sheet of metal or other material that is welded, adhered, bonded, or otherwise attached to the top surface of the platform feed structure (or mechanically coupled thereto). As such, the shape of the cover plate, in such embodiments, is a continuous or smooth surface that is a sheet that fits over and onto the top surface of the platform feed structure. In other embodiments, the cover plate can include features that aid in the turning of cooling flow as it transitions from an axial flow to a radial flow and thus reduce pressure losses associated with turning of flow that is directed into an airfoil or baffle of the vane assembly.

(45) Turning now to FIGS. **7**A-**7**C, schematic illustrations of a vane assembly **700** in accordance with an embodiment of the present disclosure is shown. The vane assembly **700** may be similar to

that shown and described above. For example, the vane assembly **700** includes an outer diameter platform **702**, an inner diameter platform **704**, and an airfoil **706** extending radially between the outer diameter platform **702** and the inner diameter platform **704**. The outer diameter platform **702** includes a forward rail **708** and an aft rail **710**. The forward rail **708** has an interior rail surface outlet **432** that directs cooling air into the baffle **412** and the outer diameter platform **702** has an interior platform surface **714**. The interior rail surface **712**, the interior platform surface **714**, and an interior rail surface of the aft rail **710** define an outer diameter plenum **716**, similar to that shown and described above.

(46) In this embodiment, the vane assembly **700** includes two airfoils, the airfoil **706** illustratively shown having a forward leading edge cavity and an aft cavity **718**. A second airfoil is arranged adjacent the first airfoil **706** and has an aft cavity **720** illustratively shown. Cooling flow may enter the aft cavities **718**, **720** of the airfoils **706** through openings that are exposed to the outer diameter plenum **716**. The cooling flow that is directed into the aft cavities **718**, **720** may not be of sufficient temperature or pressure for cooling leading edge cavities of the airfoils **706**. Accordingly, the vane assembly **700** includes platform feed structures **722**, **724** arranged radially outward from the leading edge cavities of the airfoils **706**. The platform feed structures **722**, **724** are configured to direct a cooling flow of sufficient pressure and/or temperature into leading edge cavities of the airfoils **706**.

(47) The radially oriented platform feed structures **722**, **724** are configured as shown and described above, having inlets arranged on and through the leading edge forward rail **708** and cover plates **726**, **728** that are configured to direct and turn an axial flow to a radial flow into the leading edge cavities. In this illustrative embodiment, the platform feed structures **722**, **724** have concave top surfaces (e.g., as described above) and the cover plates **726**, **728** have similar curvature to ensure a seal between the cover plates **726**, **728** and the top surfaces of the platform feed structures **722**, **724**.

(48) Referring to FIGS. 7B-7C, schematic illustrations of the cover plate **726** are shown, with the other cover plate **728** being substantially similar thereto. The illustrations of FIGS. 7B-7C are of the cover plate **726** in isolation without other features of the radially oriented platform feed structures **722**, **724** shown. As shown in FIGS. 7B-7C, the cover plate **726** includes a mounting surface **730** that is shaped to overlay a top surface of a respective platform feed structure **722**, **724** such that the mounting surface **730** may be bonded, adhered, welded, mechanically affixed, or otherwise attached in a sealing manner to the top surface of the respective platform feed structure **722**, **724**.

(49) The cover plate **726** includes a turning contour surface **732** that is shaped to turn an airflow from an axial flow direction to a radial flow direction. For example, and as shown, the turning contour surface **732** defines a dome-like shape that extends outward from the mounting surface **730**. As shown in FIG. 7B, an apex **734** of the turning contour surface **732** is configured to align with both an inlet **736** of the platform feed structure **722** in the forward rail **708** and an outlet **738** of the platform feed structure **722** that directs cooling air into a baffle installed within a leading edge cavity of the airfoil **706**, as described above.

(50) The cover plate **726**, as shown, is formed into a shell or dome shape to provide a cushion for high Mach number flow and turn such flow 90° from an axial flow direction to a radial flow direction into the airfoil or a baffle in the airfoil. Such turning contour surface **732** allows for more variety in the cooling circuit used in the leading edge of the airfoil without suffering extreme pressure losses due to turns. That is, the turning contour surface **732** provides for a smoother turning transition as compared to prior configurations. The height of the cover plate “dome” (i.e., the position of the apex **734**) can be adjusted to align with the feed height of the flow (e.g., the inlet **736**) and can vary in radius to allow for turning the flow more quickly or over a larger area to ensure uniform pressure distribution into the cavity of the airfoil (leading edge cavity and/or baffle cavity). The cover plate **726** can have a circumferential span **740** that can be selected to cover

additional or larger platform feed structures (e.g., a second feed path that extends between an inlet in the forward rail and an outlet that directs cooling air into a platform cooling cavity, as shown in FIGS. **4**A-**4**C).
(51) In some embodiments, the platform feed structures may be configured to provide feed supply of cooling air into not just the leading edge cavity (and/or the platform cooling). For example, in some embodiments, the structure of the radially oriented platform feed structures **722**, **724** shown in FIG. **7**A may extend aft to also cover or include the aft cavities **718**, **720**, such as a mid-body and/or trailing edge cavity of the airfoil (including, for example, baffles therein). As such, embodiments of the present disclosure are not limited to just supplying cooling air into leading edge cavities/baffles of airfoils but can extend aftward to cover and supply air into aft cavities such as mid-body cavities and/or trailing edge cavities.
(52) Referring now to FIGS. **8**A-**8**B, schematic illustrations of examples of a vane assembly **800** in accordance with an embodiment of the present disclosure are shown. The vane assembly **800** includes an airfoil **801** and a forward rail **802**. The forward rail **802** may be located within a compressor section or a turbine section of a gas turbine engine, as shown and described above. The forward rail **802** may operate to position and constrain the vane assembly **800** in an engine static structure. The forward rail **802** includes a radially oriented platform feed structure **804**. The radially oriented platform feed structure **804** has a first radially oriented extension **806** and a second radially oriented extension **808**. Each of the first radially oriented extension **806** and the second radially oriented extension **808** extends from an interior platform surface **810** of a platform **812**. The first radially oriented extension **806** extends from the interior platform surface **810** of the platform **812** to define a first height **814** and the second radially oriented extension **808** extends from the interior platform surface **810** of the platform **812** to define a second height **816**. In this example, the first height **814** and the second height **816** are not of heights that are substantially equal to one another. In such configurations, a pair of cover plates (not shown) may be mounted to top surfaces of the first radially oriented extension **806** and/or the second radially oriented extension **808** to close off a respective cavity defined therein.
(53) The first radially oriented extension **806** includes a portion of an outer diameter platform supply cavity **818** and the second radially oriented extension **808** includes a portion of an airfoil leading edge supply cavity **820**. It is to be understood that the term “outer diameter platform” as used herein refers to a platform located further away from an engine central longitudinal axis, such as the engine central longitudinal axis A, in comparison to an inner diameter platform. The outer diameter platform supply cavity **818** and the airfoil leading edge supply cavity **820** of the platform feed structure **804** are arranged, at least partially, within the forward rail **802** to assist in routing cooling air, as represented by arrow **822**, from a location forward of the leading edge forward rail **802** into the outer diameter platform supply cavity **818** and the airfoil leading edge supply cavity **820**. In this example, the outer diameter platform supply cavity **818** and the airfoil leading edge supply cavity **820** are in fluid communication with one another. In other embodiments, the two supply cavities **818**, **820** may be fluidly separate from each other.
(54) In some embodiments, and as shown in FIGS. **8**A-**8**B, the radially oriented platform feed structure **804** may include a first inlet **824** and a second inlet **826** formed in the forward rail **802**. The first inlet **824** is open to the outer diameter platform supply cavity **818** and the second inlet **826** is open to the airfoil leading edge supply cavity **820**. Each of the inlets **824**, **826** is open to a respective cavity and allows cooling air (e.g., the cooling air represented by arrow **822**) to flow directly into a respective cavity. The outer diameter platform supply cavity **818** is configured to direct a cooling flow into a platform cooling cavity. The airfoil leading edge supply cavity **820** is configured to direct a cooling flow into a leading edge cavity of the airfoil **801**. In this embodiment, the airfoil **801** may not include a baffle installed therein. As such, in this embodiment, the cooling flow is directed directly into a leading edge cavity of the airfoil **801**. In other configurations, the outer diameter platform supply cavity **818** and the airfoil leading edge supply

cavity **820** may be fluidly coupled and supplied with a single source of cooling air (e.g., through the second inlet **826**—being the only inlet in such a configuration).

(55) Turning now to FIG. **9**, a schematic illustration of a vane assembly **900** in accordance with an embodiment of the present disclosure is shown. The vane assembly **900** includes a platform **902** with an airfoil **904** extending radially inward therefrom and a forward rail **906**. The platform **902** includes a platform feed structure **908** arranged thereon against the forward rail **906**. The platform feed structure **908** has a first extension **910** and a second extension **912**. In this embodiment, the first extension **910** is a structure having an internal cavity configured to direct cooling flow to a platform cooling cavity **914** and the second extension **912** is a structure having an internal cavity configured to direct cooling flow to a leading edge cavity **916** of the airfoil **904**.

(56) In this embodiment, a first cover plate **918** is attached to cover a top of the first extension **910** and a second cover plate **920** is attached to cover a top of the second extension **912**. The cover plates **918**, **920** may be mounted to top surfaces of the respective first and second extensions **918**, **920** to close off respective cavities defined therein. In some embodiments, the cover plates **918**, **920** may be similar to that shown and described with respect to FIGS. **7**A-**7**C, including respective turning contour surfaces **922**, **924** that can aid in efficiently turning an axial direction cooling flow to a radial direction cooling flow into the respective platform cooling cavity **914** and leading edge cavity **916**. In this embodiment, the airfoil **904** does not include a baffle within the leading edge cavity **916**. Accordingly, a top opening of the platform **902** about the airfoil **904** is not required to be enlarged to enable installation of such baffle. As a result, the axial profile of the platform feed structure **908** (i.e., length axial distance from the forward rail **906**) may be less than the configurations shown and described with respect to FIGS. **4**A-**6**, for example. For example, because a baffle is not required to be accommodated, an axial length or extent of the platform feed structure **908** may be less than 20% of an axial length of the platform (e.g., distance between a forward rail and an aft rail of the platform).

(57) Advantageously, embodiments of the present disclosure allow for more variety in cooling configuration options for the leading edge of airfoils of vane assemblies. In accordance with some embodiments of the present disclosure, the axial extent of platform feed structures has been increased as compared to prior configurations for improved cooling in both the axial and circumferential directions. The enlarged platform feed structures allow flow more room to make a 90° turn required by the feed system, thus reducing pressure losses and attempting to simulate a plenum for more uniform distribution of the flow into the cavity of a baffle or airfoil. Additionally, prior platform feed structures were subject to plugged feed holes due to particulates (e.g., dirt). However, advantageously, the platform feed structures permit relatively large feed slots that minimize the risk of plugged feed holes, thus reducing the risk of starving the entire circuit (which leads to burning).

(58) Further, advantageously, in accordance with some embodiments, improved cover plates for platform feed structures are provided. The geometric features of the cover plates, in combination with the platform feed structures, allows a flow to be introduced into a cooling circuit in a controlled, predictable, and uniform capacity at varying levels of Mach number with minimized pressure loss. The more uniform distribution allows flexibility in where and how the flow is introduced and used within the airfoil itself. Previous technologies utilize flat cover plates which can cause unpredictability and non-uniform pressure distributions depending on the Mach number and feed size being used. Without the contoured cover plates described herein, it becomes more difficult to simulate a plenum distribution with higher Mach number flow.

(59) The use of the terms “a”, “an”, “the”, and similar references in the context of description (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or specifically contradicted by context. The modifier “about” used in connection with a quantity is inclusive of the stated value and has the meaning dictated by the context (e.g., it includes the degree of error associated with measurement of the

particular quantity). All ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. As used herein, the terms “about” and “substantially” are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, the terms may include a range of ±8%, or 5%, or 2% of a given value or other percentage change as will be appreciated by those of skill in the art for the particular measurement and/or dimensions referred to herein. It should be appreciated that relative positional terms such as “forward,” “aft,” “upper,” “lower,” “above,” “below,” and the like are with reference to normal operational attitude and should not be considered otherwise limiting.
(60) While the present disclosure has been described with reference to an example embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A vane assembly for a gas turbine engine, the vane assembly comprising: a platform having an interior platform surface, a forward rail, and an aft rail, wherein the interior platform surface, the forward rail, and the aft rail define a plenum; an airfoil extending radially inward from the platform on a side opposite the forward and aft rails, the airfoil having a leading edge cavity that defines an opening in the platform; a platform feed structure arranged on the platform about the opening of the leading edge cavity and arranged in the plenum and defining a fluid path through an inlet in the forward rail and into an outlet in the platform that is open to the leading edge cavity of the airfoil; and a cover plate arranged on a top surface of the platform feed structure, the cover plate configured to fluidly separate the plenum of the platform from the leading edge cavity of the airfoil and define a turning plenum, wherein the cover plate comprises a mounting surface for mounting to the top surface of the platform feed structure and a turning contour surface that extends from the mounting surface and is dome-shaped to turn an airflow from an axial flow direction to a radial flow direction within the turning plenum; wherein the turning contour defines an apex on the cover plate that is aligned with each of the inlet and the outlet of the platform feed structure.
2. The vane assembly of claim 1, wherein the platform feed structure has an open top that is defined by the top surface and the cover plate is attached to the top surface.
3. The vane assembly of claim 2, wherein the cover plate is attached to the top surface by one of welding, bonding, and adhesive.
4. The vane assembly of claim 1, wherein the top surface is substantially planar extending from a forward position on an interior rail surface of the forward rail and an aft position on the interior platform surface of the platform.
5. The vane assembly of claim 1, wherein the outlet is aligned with the leading edge cavity of the airfoil.
6. The vane assembly of claim 1, wherein the top surface defines a contoured surface extending from a forward position on an interior rail surface of the forward rail and an aft position on the interior platform surface of the platform.
7. The vane assembly of claim 6, wherein the contoured surface is convex.
8. The vane assembly of claim 6, wherein the contoured surface is concave.
9. The vane assembly of claim 1, wherein the platform feed structure defines a first feed path through the platform feed structure from the forward rail to the leading edge cavity and a second

feed path through the platform feed structure from the forward rail to a platform cooling cavity of the platform.

10. The vane assembly of claim 9, wherein the first feed path is defined through a first extension of the platform feed structure and the second feed path is defined through a second extension of the platform feed structure, wherein the cover plate is positioned over a portion of at least the first feed path.

11. The vane assembly of claim 10, wherein the first extension and the second extension have different heights relative to the interior platform surface.

12. The vane assembly of claim 1, wherein the platform feed structure extends at least 20% of an axial distance between the forward rail and the aft rail.

13. The vane assembly of claim 1, wherein the platform feed structure extends between 20% and 70% of an axial distance between the forward rail and the aft rail.

14. The vane assembly of claim 1, wherein the platform feed structure extends less than 20% of an axial distance between the forward rail and the aft rail.

15. The vane assembly of claim 1, wherein the airfoil comprises an aft cavity arranged aft of the leading edge cavity, wherein the aft cavity is fluidly connected to the plenum of the platform.

16. The vane assembly of claim 1, wherein the platform is an outer diameter platform and the airfoil extends radially inward to an inner diameter platform.

17. The vane assembly of claim 1, wherein the platform feed structure is sized to allow a baffle to pass therethrough to install the baffle within the leading edge cavity of the airfoil.